

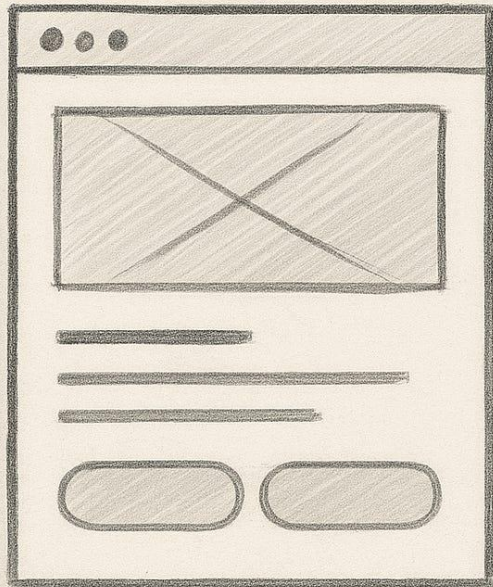
CSE472: HCI – Human Computer Interaction

Lecture – 11: Hi-Fidelity Prototyping

Spring 2026



Inspiring Excellence



Low-Fi



High-Fi

<https://uxuniversity.io/p/how-to-prototype-faster-without-sacrificing>

What is Prototype?

- A prototype is an incomplete, early version of a product
- There are many approaches to building prototypes for software user interfaces
- UI prototypes can be as simple as a drawing on a piece of paper or as complex as functional web application
- Or anywhere in between those extremes!



Lo~fi vs. Hi~fi

- Traditionally, prototypes are categorized as either lo-fi (low fidelity) or hi-fi (high fidelity)
- Fidelity can be thought of as how close of an approximation of the final product is being attempted
- In this class, we will often use lo-fi as a synonym for paper prototyping and hi-fi as a synonym for digital prototyping
- Your design process should use multiple levels of fidelity as you move from an idea to a fully designed product

Dimensions of Fidelity

Fidelity can be broken down into four basic dimensions:

- Breadth
- Depth
- Look
- Interaction

Breadth

The “breadth” of a prototype refers to how much of the product’s functionality is represented in the prototype

- A very narrow prototype only represents a single feature
- A broad prototype represents all intended functionality
- Prototypes should generally be as broad as needed to cover basic or most important tasks, but not much more

Depth

The “depth” of a prototype refers to how much of the prototype is functional, and how robust it is

- A very **shallow prototype** has no backend at all and is hard-coded to respond as though the user had provided ideal input
- A **deep prototype** has some logic and error-handling capabilities
- At first glance, depth may seem unimportant, but it affects the **amount of exploration** a user can do
- Thus depth can actually have a **profound influence** on user testing!

Look

“Look” is probably what most people think of when they think of prototype fidelity

- It refers to how accurately a prototype represents the product’s intended appearance, including fonts, colors, and graphics
- It’s generally a good idea to hold off on something that has a high fidelity look until later in the design process
 - People are less likely to point out flaws and mistakes
 - People can easily fixate on the “little” things
 - You are less likely to throw it out and start again

Interaction

“Interaction” refers to how the prototype handles input and output

- Interaction can often be simulated
- For example, you might create a digital prototype for an iPad application which runs on your desktop and responds to a traditional mouse and keyboard
- You might use hyperlinks or animation to simulate clicking interaction (e.g., in Powerpoint)

Hi-fi Prototyping

- Once you have developed a lo-fi prototype and solicited feedback on it through peer critique and user testing:
 - You may wish to create another lo-fi prototype
 - (isn't iterative design fun?)
 - Or you may be ready to move on to a hi-fi prototype
- Remember, a high fidelity prototype is a substantial time investment!
- It is good for evaluating a working design that has been derived from a few rounds paper prototyping

Things you will need to consider

- Choice of tool (more in a minute)
 - Typography / font
 - Color palette
 - Device
 - Interaction
 - Implementation
- Start by creating a digital version of your paper prototype
- Then iterate through user testing and feedback
- *Don't design a beautiful prototype that can't be implemented!*

Digital Prototyping Tools

- There are literally hundreds...
- And more released every day.
- *I don't know them all!*
- What you choose will depend on a variety of factors...

Choosing a tool: Considerations

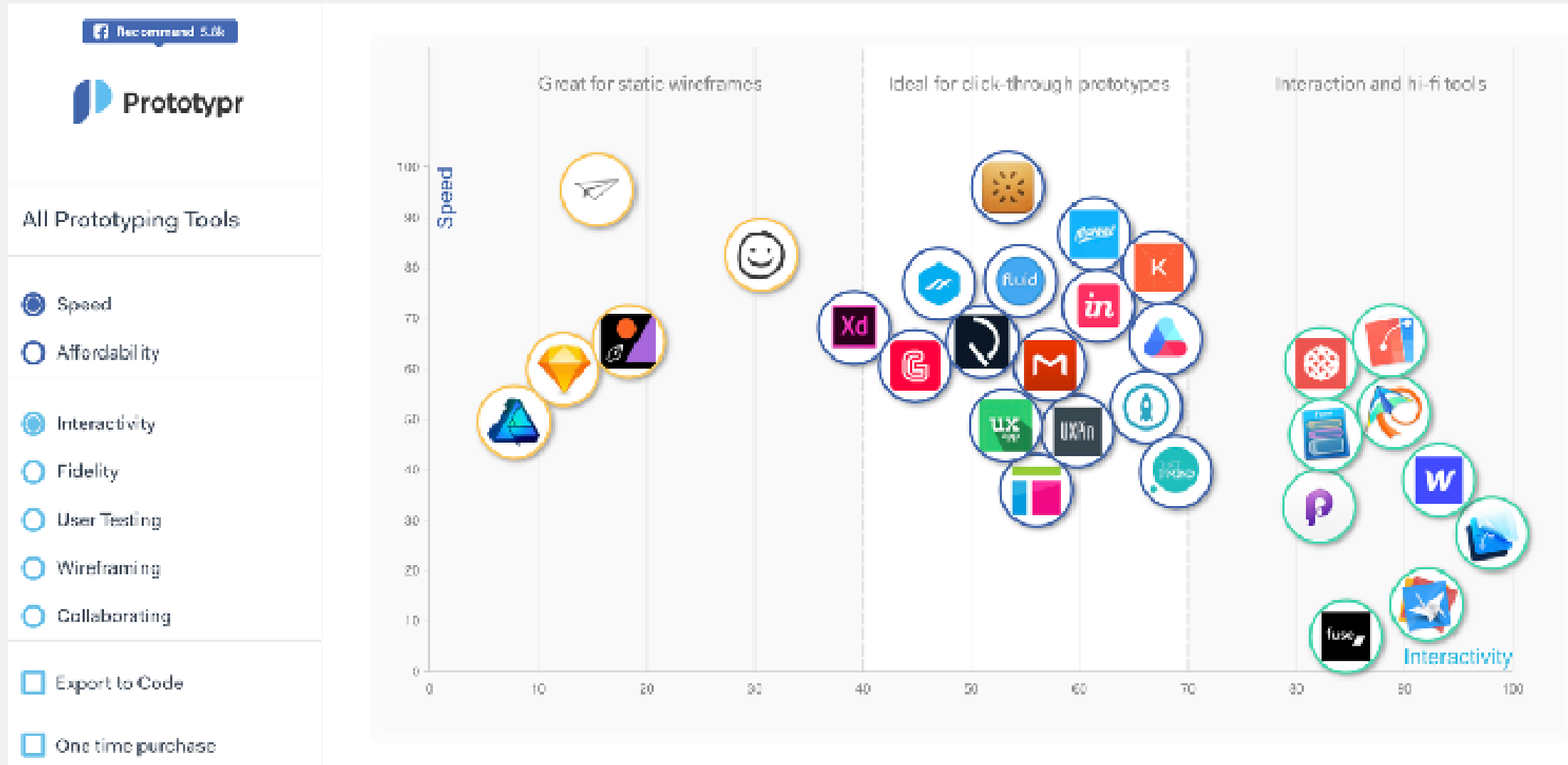
- Learning Curve
 - How long will it take me to learn this tool?
- Usage
 - Which device will it be used on?
- Fidelity
 - Will it showcase layout structure or complex interactions?
- Sharing
 - Can I collaborate with others on the prototype?
- Cost
 - How much am I prepared to pay for this tool?

Comparing prototyping tools

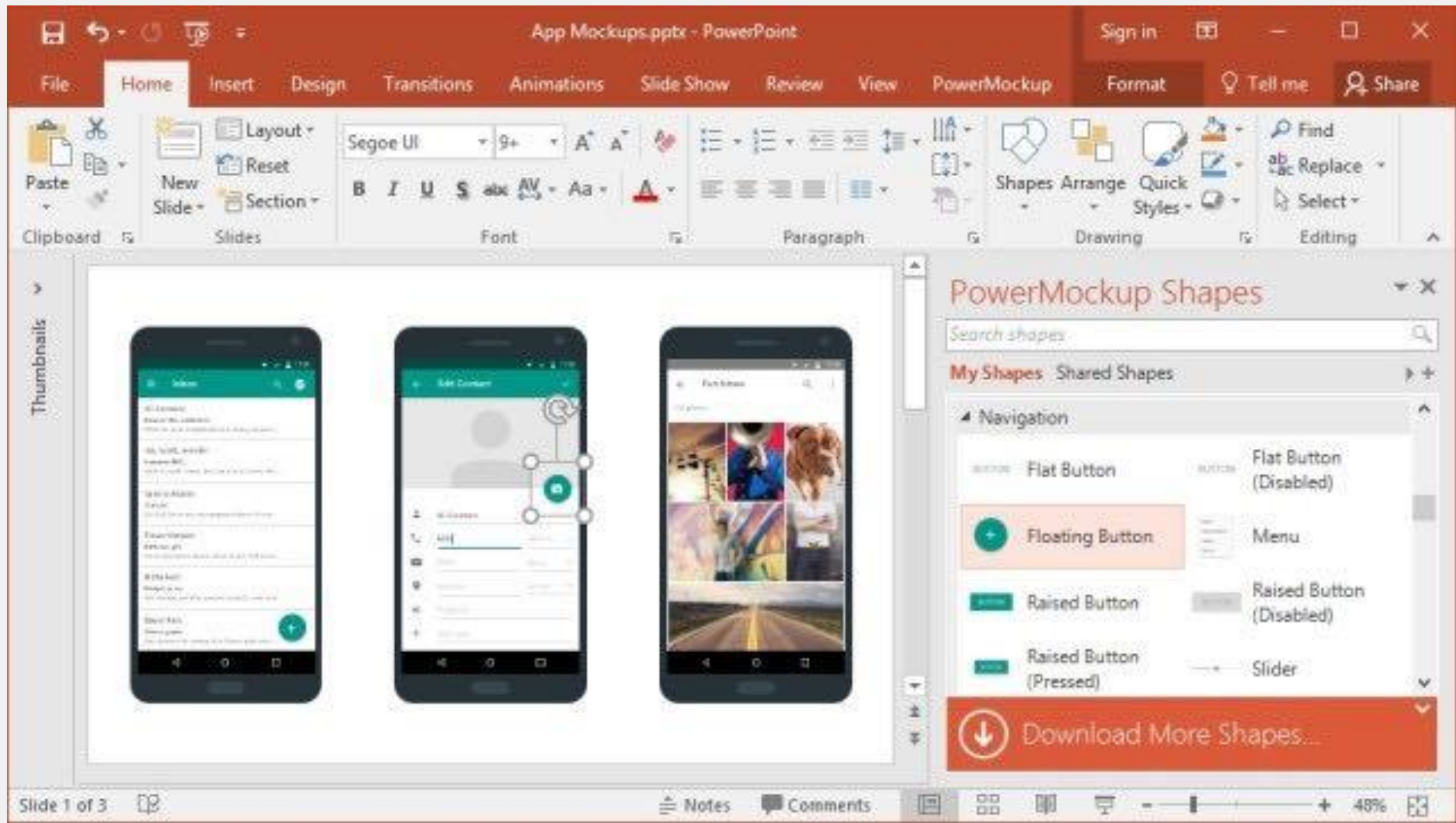
- A cool tool to help you do this:

<http://www.prototypr.io/prototyping-tools/>

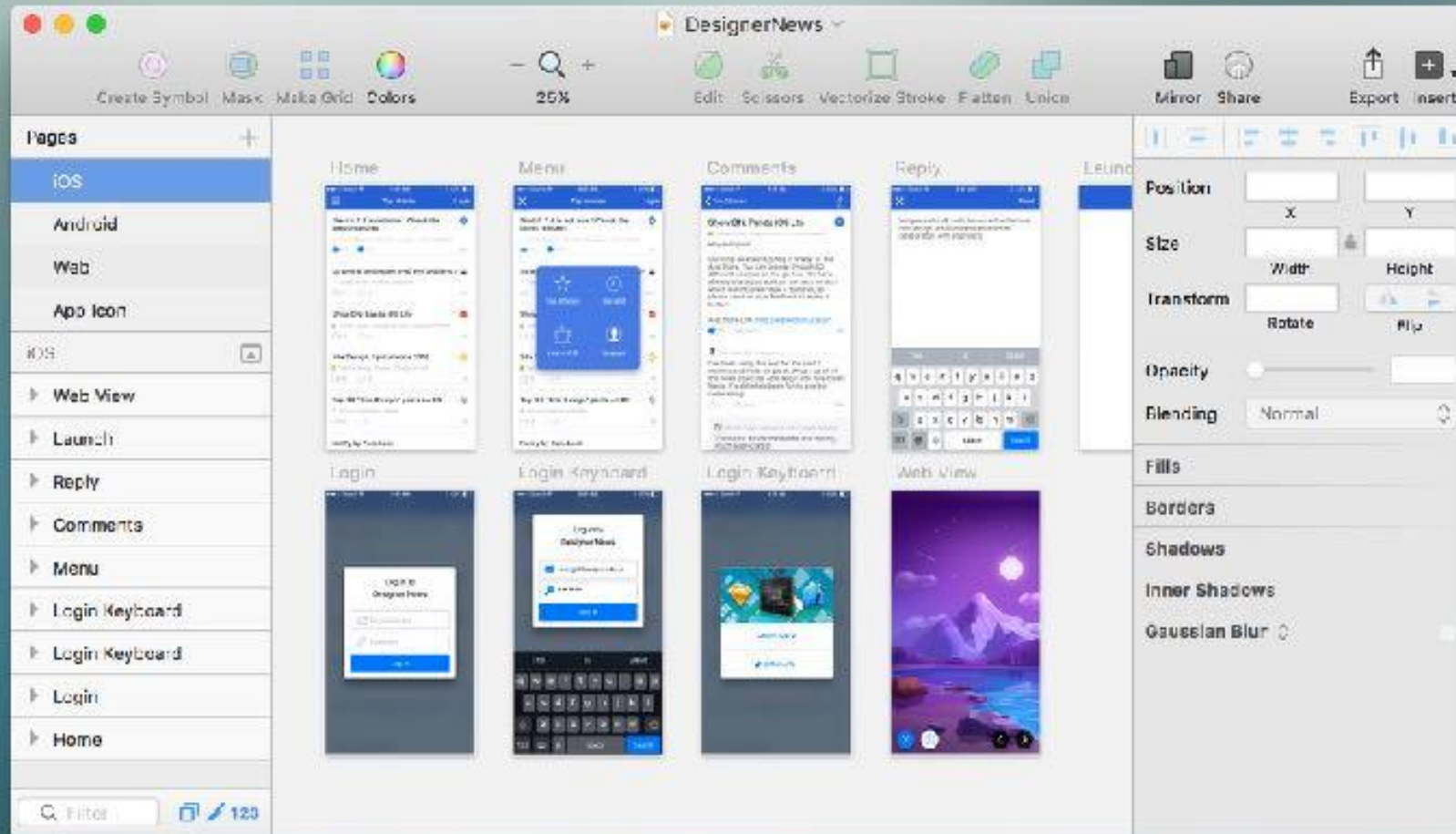
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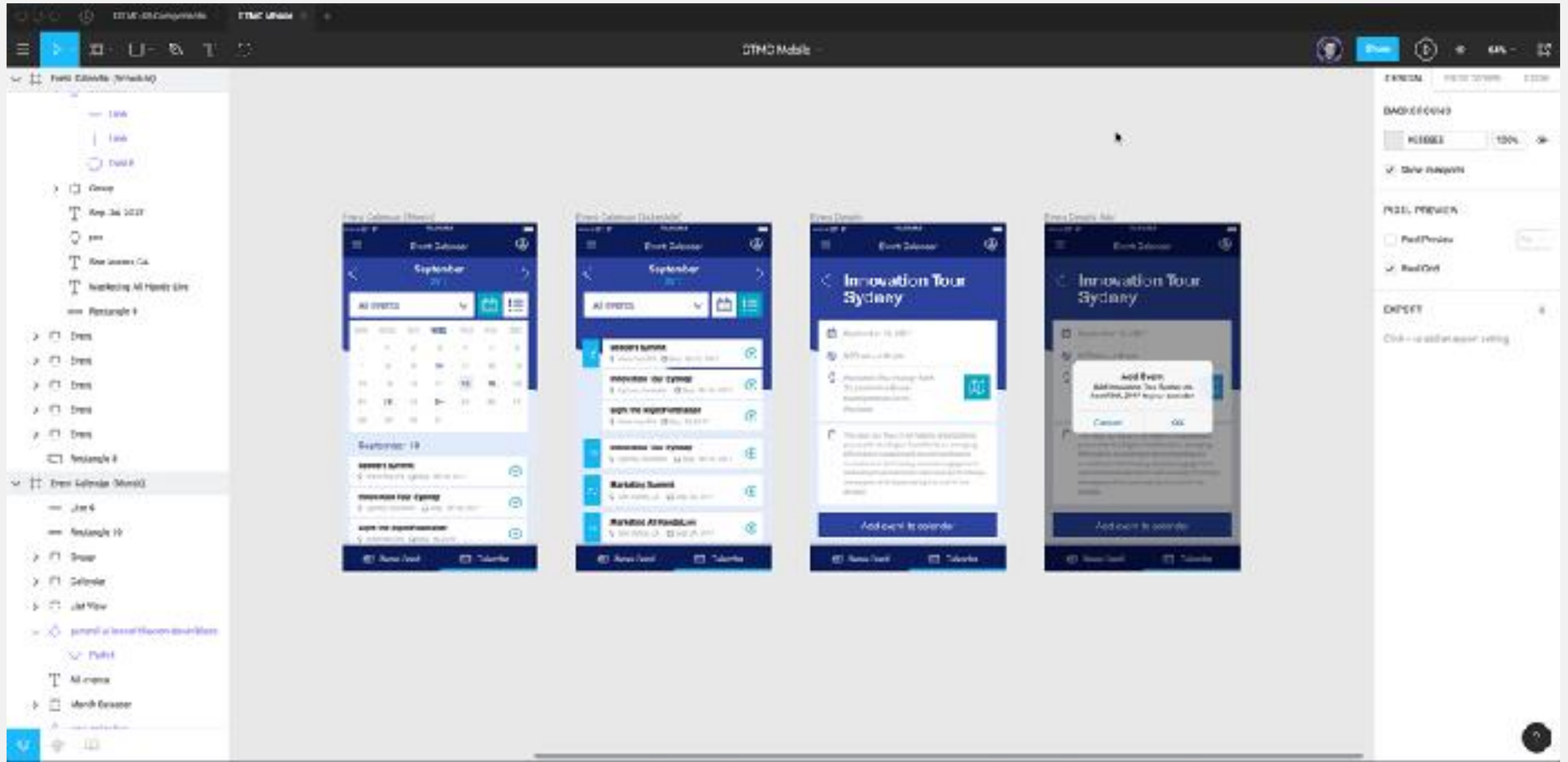
Powerpoint or Keynote



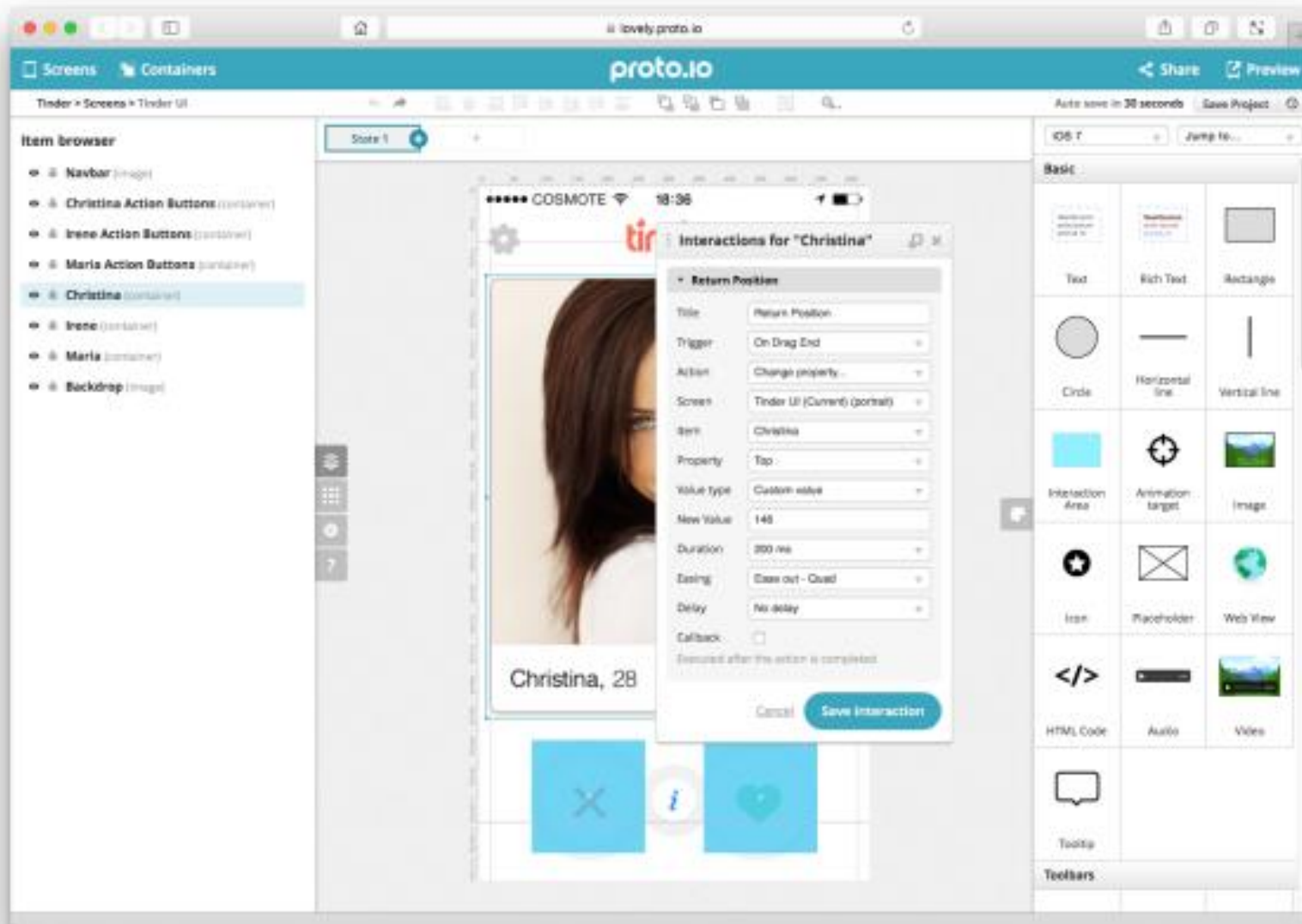
Sketch



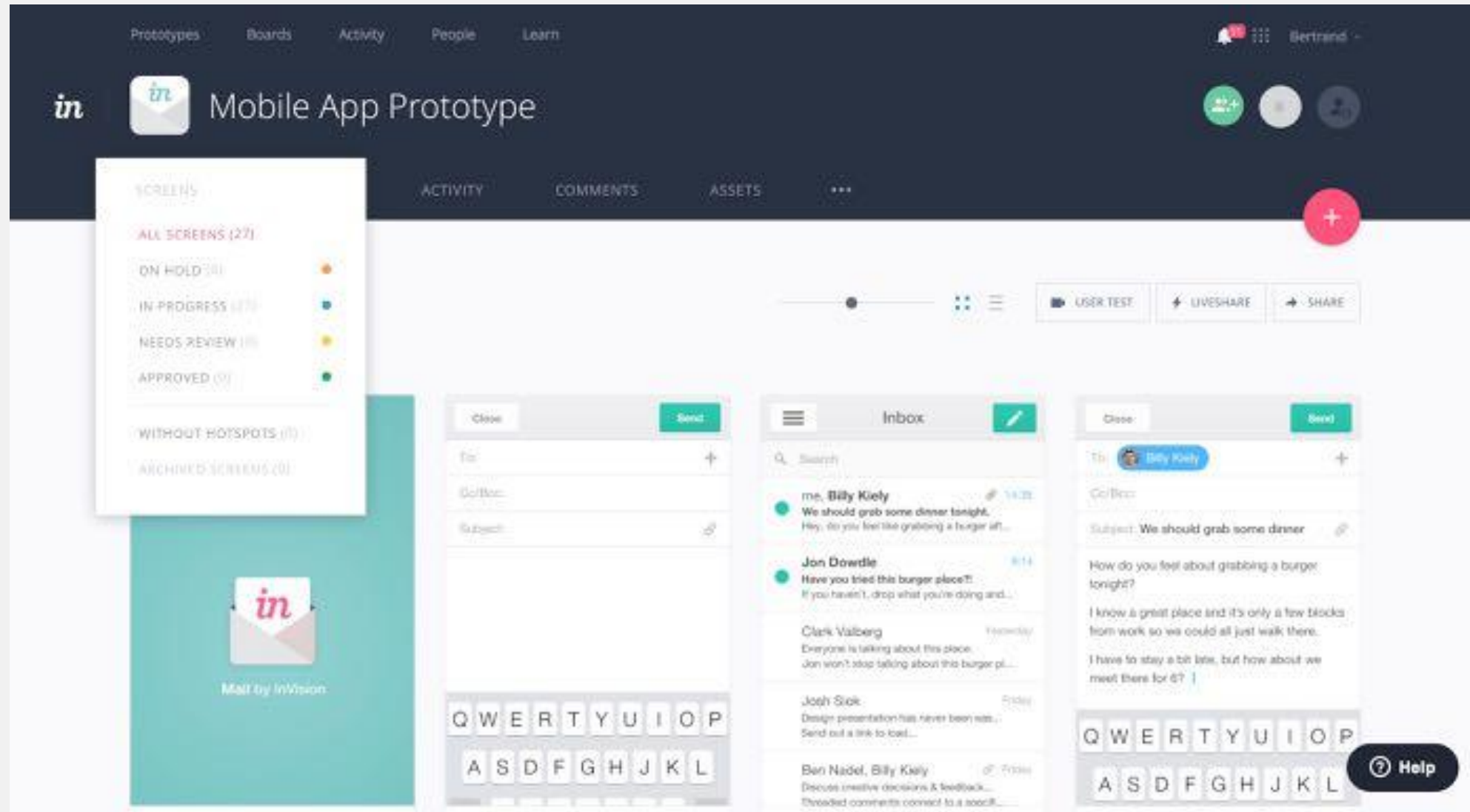
Figma



Proto.io



InVision



Any questions?